Question:

A group of scientists is studying the mechanism by which the human papillomavirus (HPV) vaccine confers immunity. They observe that during the inoculation of test subjects, mammals with certain viral proteins result in the organism’s antigen-presenting cells (APCs) absorbing the antigen and presenting it on major histocompatibility complex (MHC) class 1 molecules. Which of the following is the correct term for the process that the scientists are observing in this inoculation?Answer Choices:A. Endogenous antigen presentationB. Cross-presentationC. Priming of CD4+ T cellsD. Adhesion

Reasoning + Answer:

<think>The process being observed by the scientists is the presentation of antigens to the immune system. The antigens are viral proteins that are introduced to the organism, which are then absorbed by antigen-presenting cells (APCs). These APCs process the antigens and present them on major histocompatibility complex (MHC) class 1 molecules. This is an example of exogenous antigen presentation, but the specific process of presenting antigens from outside the cells, such as viruses or bacteria, to cytotoxic T cells (which recognize antigens presented on MHC class 1 molecules) is known as cross-presentation. This is because the antigens are being presented to T cells that do not typically interact with antigens presented on MHC class 1 molecules.</think><answer>B. Cross-presentation</answer>